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Claude Certified Architect Foundations · Anthropic · 2026 · [Verified on Credly](#)

Summary

Senior software engineer with a focus on LLM systems and healthcare innovation. Looking for a team where I can contribute AI expertise and continue a track record of mentoring engineers and building healthy team practices.

Featured Project: Meno

Status: In active development, 2021–present · independent, self-funded

What it is: Full-stack healthcare app helping women navigate menopause care: a NAMS-provider directory with call-tracking, RAG-powered clinical Q&A, and an appointment-prep flow that turns symptom logs into a provider-ready clinical summary.

- **RAG pipeline:** pgvector with an hnsw index across 10,000+ curated research documents; five-layer prompt system governs voice, citation format, and medical-safety boundaries; responses cite sources at the paragraph level.
- **Structured LLM pipeline:** Six-call appointment-prep flow (narrative generation, symptom summary, provider questions, RAG-backed scenario suggestions) with Pydantic-validated structured outputs; hard-fails on parse errors rather than returning partial output.
- **Provider-agnostic architecture:** Built on the OpenAI API with dependency injection throughout; swapping to the Claude API for production requires changing one file.
- **Stack:** SvelteKit + TypeScript, FastAPI + Python, Supabase (PostgreSQL + pgvector, row-level security), Claude API / OpenAI API

Experience

Flashpoint | Remote (Vancouver, WA)

Nov 2019 – Oct 2021

Senior Software Engineer / Technical Lead

- **Compromised Credentials Monitoring (CCM-E):** Built real-time breach detection system alerting enterprises when employee credentials appeared in compromised datasets. Unified 6+ data sources into queryable system delivering 35B+ records via API, UI, and alerts.
- **Data Warehouse Cleansing:** Led effort to validate, enrich, and optimize 40TB raw dataset. Designed schema improvements enabling faster searches and new customer insights.

Recall Masters | Remote (Vancouver, WA)

May 2019 – Sep 2019

Senior Software Engineer / Technical Lead / Supervisor

- **Team Leadership:** Mentored engineers in best practices, code quality, testing, and healthy team dynamics.
- **Modernization:** Converted legacy Python monolith to modern architecture with Lambda microservices.
- **Data Integration:** Built Elasticsearch warehouse connecting siloed dealer data sources for unified reporting.

Cox Automotive | Remote (Vancouver, WA)

May 2017 – May 2019

Senior Software Engineer – Data Solutions

- **Vehicle Safety Recalls:** Led data integration uniting 20+ APIs to help Maryland DMV identify unrepaired recall vehicles at registration. Designed Python/Pandas/PostgreSQL pipeline with resilient AWS infrastructure. Shipped 2 weeks early.
- **Vehicle Valuations AI:** Replaced legacy valuation logic with deep learning model (TensorFlow/Keras) improving accuracy and enabling standardized pricing across brands.

Software Engineer

Worked on highly-available async API (Twisted/Python). Built security forensics analytics tool (Elasticsearch ELK, C#/.NET).

Intel | Hillsboro, OR

2012 – 2016

Software Engineer

Started as intern, worked across data/analytics. Mapped Intel's systems and infrastructure dependencies as a graph (Neo4J, Elasticsearch), primarily serving the BCDR (Business Continuity/Disaster Recovery) team's continuity and recovery planning across the division. Broad experience with Hadoop, Hive, Cassandra.

Agentic Engineering Practice

Disciplined seven-stage agentic engineering workflow: PRD-first design, TDD with CATCHES-annotated tests (every test documents the bug it prevents), Claude Code with custom architectural conventions enforced consistently, and line-by-line human review before every merge. Full methodology at helenalucia.vercel.app/process.

Technical Skills & Expertise

AI/LLM Systems

Claude API • Claude Code • RAG pipelines • Vector embeddings • Prompt engineering • MCP

Backend & APIs

Python 3.9+ • FastAPI • Flask • REST/SOAP

Data & Analytics

Pandas • NumPy • PostgreSQL • Elasticsearch • TensorFlow • Plotly

Frontend

SvelteKit • TypeScript • HTML/CSS

Cloud & Infrastructure

AWS • GCP • Supabase • Docker • Kubernetes • Terraform

Databases

PostgreSQL • Elasticsearch • Neo4J • Hadoop

Learning & Certifications (2025–2026)

Coursera AI Specializations & Courses

Intensive, hands-on learning across two specializations and 18 courses. Focus: LLM architectures, agentic AI, prompt engineering, Claude API integration, responsible AI development.

Specializations: Generative AI Software Engineering (Vanderbilt) • AI Agent Developer (Vanderbilt)

Selected Coursework: Building with the Claude API • Claude Code in Action • Introduction to Model Context Protocol • AI Fundamentals with Claude

Education

Washington State University | Vancouver, WA

2014

Bachelor of Science, Computer Science | Cum Laude

Chancellor's Award • Grace Hopper Scholar • Dean's List • ACM-W Vice Chair